



WACE Examination, 2008

ENGINEERING STUDIES

Section 2, Materials, Structures and Mechanical Systems

STAGE 3

Provisional Marking Guidelines

Section 2

Part A: Multiple Choice Questions

1	A
2	B
3	D
4	C
5	D
6	B
7	B
8	C
9	C
10	A
	[1 mark each]

Part B: Written (Stage 3)

Q1

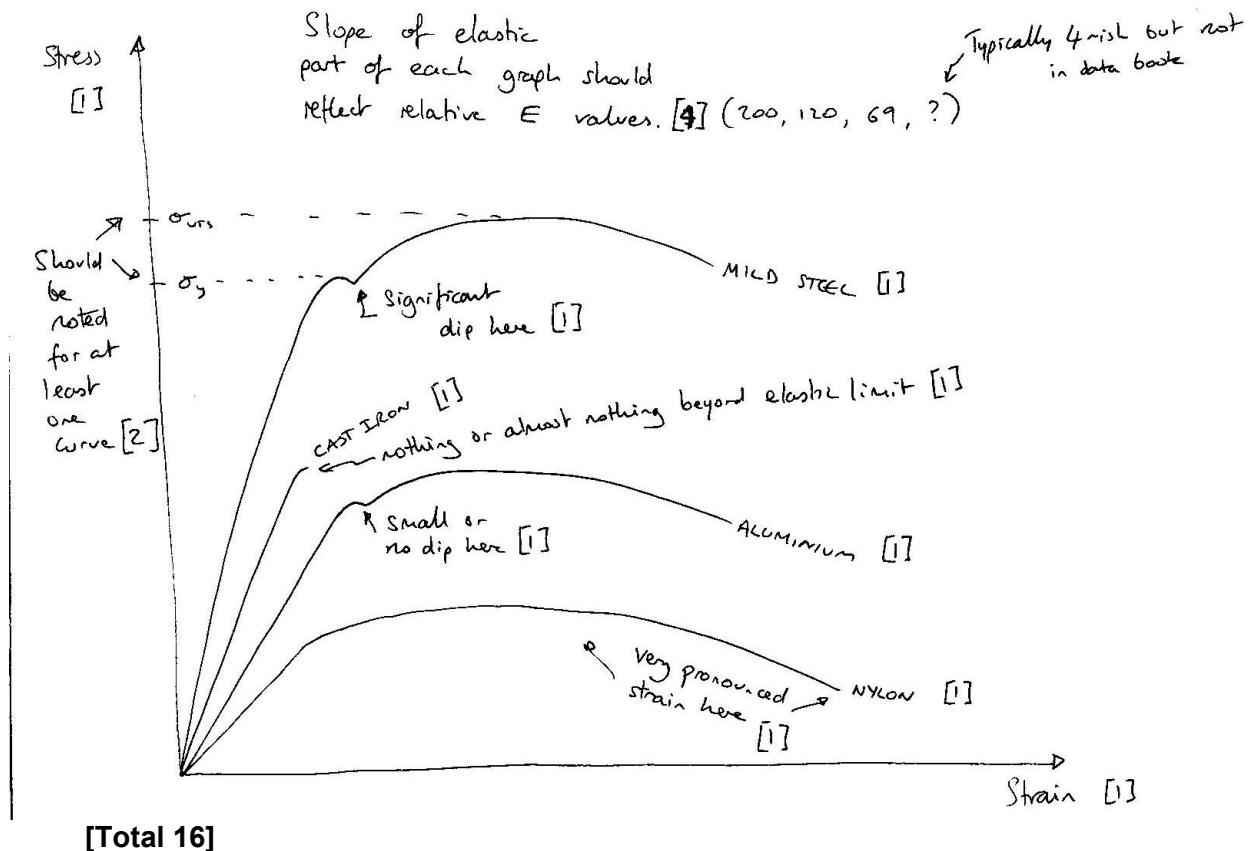
- (a) Worm does 2000rpm;
40 revolutions of worm give one turn of the 80/40 wheel; [2]
 $2000/40 = 50\text{rpm}$ [2]
120 wheel will turn at $80/120 \times 50\text{rpm} = 33.33\text{rpm}$ [2]
[Total 6]
- (b) Circumference of drum = $\pi d = 3.142 \times 0.2 = 0.628\text{m}$ [2]
Drum turns at 33.33rpm; in 50 seconds turns $33.33 \times 50/60 = 27.78$ turns [2]
Distance is therefore $27.78 \times 0.628 = 17.44\text{m}$ [2]
[Total 6]
- (c) Many different ways of expressing torque, e.g. "Rotational force" or "Turning force"
analogous to Force in non-rotating systems; Force times lever arm; Force times
perpendicular distance from the pivot to the line of action of the force. (4 lines)
[Total 2]
- (d) $T = \text{force} \times \text{radius}; 600 \times 0.1 = 60 \text{ Nm}$ (1 line)
[Total 2]
- (e) Torque from part (d) is output torque; input torque from graph is 11.5N Answer between
11 and 12 is OK.
[Total 4]

Q2

- (a) (One or other of the answers below is sufficient for one mark each).
(A) Elastic limit; Proportional limit
(B) Yield point; yield
(C) Ultimate Tensile Strength; UTS; Maximum tensile stress
(D) Failure; fracture
[Total 4]
- (b) Tensile test. Specimen is held at each end [1] in the jaws of a tensometer, which applies
a steadily increasing load [1] to the specimen while recording load and extension [1].
Load and extension are used to calculate stress and strain.[1]
[Total 4]

(c)

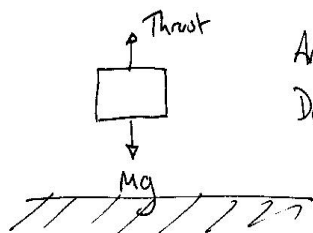
STAGE 3 Q12(c) Marking key.



Q3

(a) See hand-drawn figure. [3]

Q13 (a) STAGE 2 AND STAGE 3. Marking key



Arrows in right directions [2]

Downward force identified as Mg [1]

[Total 3]

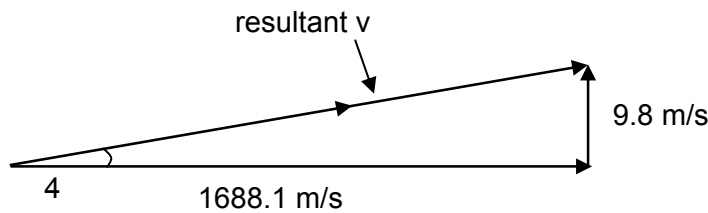
(quarter page blank)

- (b) Net force = Thrust - (mg) = $16000 - (4550 \times 1.62)$ = 8629N [2]
 $F=ma$; $a=F/m$ = $8629/4550$ = 1.87 m/s^2 [2]
After 10 seconds, $v=u+at$ ($u=0$) $v = 1.869 \times 10 = 18.69 \text{ m/s}$ [2]
 $S=ut+\frac{1}{2}at^2$; ($u=0$) $s = \frac{1}{2}at^2 = 1.869 \times 100/2 = 94.82 \text{ m}$ [2]
[1 mark for suitable rounding of answers i.e. to no more than 2 dp]
[Total 9]

- (c) $v=u+at$ again, $v = 1.869 \times (7 \times 60 + 14)$ = 822.86 m/s [2]
Acceleration will not stay constant because the mass is decreasing as the fuel is consumed. The acceleration will therefore increase as time passes, and the final velocity will be much higher. [2]

[Total 4]

- (d) $v_o = \sqrt{gr}$ [given] = $\sqrt{(1.62 \times 1737.1 \times 10^3)} = 1677.5 \text{ m/s}$
[Total 2] No marks if wrong g used or units not consistent.
- (e) See Figure below; [2] for diagram.



Velocity labels in correct places [1]
 Angle with surface identified [1]

Resultant velocity = $\sqrt{(1688.1^2 + 9.8^2)} = 1688.13 \text{ m/s}$ [2]

Angle with Moon's surface = α

$\tan \alpha = 9.8/1688.1$; so $\alpha = 0.33^\circ$ [2]

(Cos or Sin will also do, but Tan uses the velocities given, so reduces chance of error)

[Total 6] (quarter page blank plus 3 lines)

Q4

- (a) From data book, $\rho_{\text{brass}} = 8740 \text{ kg/m}^3$ [1]
 x-sectional area $A = \frac{3}{2} \sqrt{3} a^2$ where $a = 6 \text{ mm} = 0.006 \text{ m}$
 $A = 9.353 \times 10^{-5} \text{ m}^2$ [1]
 $A \times \rho = \text{mass per unit length i.e. kg/m}$
 $= 9.353 \times 10^{-5} \times 8740 = 0.817 \text{ kg/m}$ [1]
 so $\omega = 9.80 \times 0.817 = 8.0 \text{ N/m}$ [1]
[Total 4]
- (b) From data book, end deflection δ of cantilevered beam with uniform distribution of load ω
 $\delta = \omega L^4 / (8EI)$ [2]
 $\omega = 8.0$ from part (a)
 $L = 0.745 \text{ m}$ (given)
 $E = 110 \text{ kN/mm}^2$ from data book; $= 110 \times 10^9 \text{ N/m}^2$ [2]
 $I = \frac{5}{16} \sqrt{3} a^4$ (given)
 $= \frac{5}{16} \sqrt{3} 0.006^4$ ('a' converted to metres) $= 7.015 \times 10^{-10} \text{ m}^4$ [3 plus 1 for units correct]
 So $\delta = 8.0 \times 0.745^4 / (8 \cdot 110 \times 10^9 \cdot 7.015 \times 10^{-10}) = 4.00 \times 10^{-3} \text{ m} = 4 \text{ mm}$ [2 for answer, 1 for consistent units, 1 for suitable rounding]
[Total 12]
- (c) Reduce length [1] - deflection proportional to L^4 so small reduction will have large effect. [1]
 Change material [1] – higher Young's Modulus will give smaller deflection. [1]
 Increase bar diameter [1] – will increase I and therefore reduce deflection. [1]
[Total 6]